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Vitamin D Supplementation in Adults with Depressive Disorder: A Retrospective Record Review of a Patient Panel

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A B S T R A C T

This quality improvement project evaluated the impact of vitamin D supplementation on depressive symptoms in adult psychiatric outpatients with documented vitamin D deficiency and depressive disorders. A retrospective medical record review was conducted to assess changes in 9-item Patient Health Questionnaire scores after daily supplementation with 4,000 IU of vitamin D and standardized patient education. The intervention resulted in a statistically and clinically significant reduction in depressive symptoms, supporting the integration of vitamin D screening and supplementation into routine mental health care.
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Introduction

Problem Description

Depressive disorder is a complex and debilitating condition that affects an estimated 7% of the United States (US) population annually and compromises emotional well-being, physical health, interpersonal relationships, and daily functioning. Difficult to treat depression affects ~30% of individuals with major depressive disorder and costs the US >$43 billion annually.[1,2] The limitations of current pharmacologic treatments, including adverse effects, such as selective serotonin reuptake inhibitor (SSRI)—induced sexual dysfunction, which occurs in up to 80% of cases, often force patients to choose between symptom relief and maintaining their quality of life.[3]

Vitamin D supplementation has shown potential as an adjunctive or stand-alone treatment, improving depressive symptoms through serotonergic modulation, neuroinflammation reduction, and enhanced neuroplasticity.[4,5] Oral supplementation has also proven more effective than sunlight exposure in correcting deficiency,[6,7] particularly among individuals with insufficient serum vitamin D levels. Given its feasible implication in the role of depression and its favorable safety profile, vitamin D supplementation presents a promising, nonpharmacologic strategy to support mental health without compromising quality of life parameters.

Vitamin D is a fat-soluble secosteroid hormone essential for calcium homeostasis with recognized immunomodulatory and neurobiologic effects. Vitamin D status is assessed using serum 25-hydroxyvitamin D, with concentrations <20 to 30 ng/mL generally considered deficient or insufficient and levels of 30 to 50 ng/mL commonly regarded as sufficient for skeletal health, although optimal thresholds for extraskeletal outcomes remain an area of ongoing investigation.[8,9]

Vitamin D receptors and activating enzymes are widely distributed throughout the central nervous system, including regions involved in mood regulation, such as the prefrontal cortex, hippocampus, and limbic system.[10,11] These findings, coupled with evidence linking low vitamin D status to depressive symptoms, support a biologically plausible role for vitamin D in mental health and highlight its relevance as a potentially modifiable factor in clinical practice.

Available Knowledge

A recent systematic review by Borges-Vieira and Cardoso[4] evaluated 20 randomized controlled trials (n = 2,256) investigating vitamin D supplementation, alone or combined with B vitamins, in patients with depressive or anxious symptoms and major depressive disorder. They found significant reductions in depression scores with supplementation regimens using vitamin D either as monotherapy or adjunctive therapy in conjunction with SSRIs.[4,12-15] An umbrella meta-analysis by Musazadeh et al,[16] which synthesized 10 interventional and 4 observational meta-analyses involving >62,000 participants aged 28 to 73 years, identified that vitamin D supplementation (doses ranging from 2,500 to 6,000 IU daily) produced a moderate reduction in depressive symptoms and that higher serum vitamin D levels were associated with lower incidences of depression. These results suggest that addressing vitamin D insufficiency could have a

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positive impact on depression, particularly for individuals who start out with deficient levels.

Rationale

Research increasingly supports the therapeutic potential of vitamin D supplementation in reducing depressive symptoms. Studies show that using vitamin D as both monotherapy and as an adjunct with SSRIs has clinically significant improvements in mood and symptom severity across diverse populations.[4,12-23] The favorable safety profile and efficacy of vitamin D, as reported in multiple randomized controlled trials and meta-analyses, underscore its promise as a low-risk, accessible intervention for individuals with depressive disorders, particularly those with documented vitamin D deficiency.

Specific Aims

This project evaluated the impact of vitamin D supplementation, combined with standardized patient education, on depressive symptoms among adult psychiatric mental health outpatients. A retrospective medical record review was conducted using a convenience sample of patients diagnosed with both depressive disorders and vitamin D deficiency. The specific objectives were to (1) assess whether vitamin D supplementation led to a clinically meaningful reduction in depressive symptoms, defined as an average decrease of $\geq$3 points on the 9-item Patient Health Questionnaire (PHQ-9),[24] and (2) develop a standardized anticipatory guidance document for patient education, outlining the evidence-based role of vitamin D in managing depressive symptoms. This resource is intended to support clinical decision-making and enhance adherence to supplementation protocols among patients with confirmed vitamin D deficiency.

Methods

Context

This project was conducted in a private outpatient psychiatric practice located in northern New England, where environmental factors, such as long winters, limited daylight, and reduced sun exposure, contribute to widespread vitamin D deficiency. The practice serves a panel of >100 adult patients diagnosed with depressive disorders, most of whom are seen monthly. Approximately 96% of patients have private insurance, with the remaining 4% self-paying.

The target population for this retrospective medical record review included adults aged 18 to 64 years with a documented diagnosis of depressive disorder and vitamin D deficiency who had previously received pharmacologic treatment without achieving therapeutic symptom relief. Patients were excluded if they had any of the following: (1) depressive disorder with psychotic features, (2) dual diagnosis with uncontrolled substance use disorders, (3) bipolar I disorder, (4) neurocognitive disorders, (5) somatic symptom disorder, (6) absence of a confirmed depressive disorder diagnosis, or (7) medical conditions contraindicating vitamin D supplementation, including hypercalcemia, hyperparathyroidism, sarcoidosis or other granulomatous diseases, kidney disease with impaired calcium-phosphate metabolism, or concurrent use of medications known to interact adversely with vitamin D (eg, thiazide diuretics, digoxin).

Intervention

The intervention consisted of daily oral supplementation with 4,000 IU of vitamin D, consistent with recommendations from the Clinician Guidelines for the Treatment of Psychiatric Disorders with Nutraceuticals and Phytoceuticals.[25] Supplementation was initiated in patients with serum 25-hydroxyvitamin D levels <50 ng/mL, a threshold supported by emerging evidence suggesting that neuropsychiatric and immunomodulatory benefits of vitamin D may occur at levels higher than those required solely for bone health.[9] All participants received a standardized educational resource outlining the evidence-based role of vitamin D in mood regulation and depression management. This educational component was designed to enhance adherence and support informed decision-making.

Step 1

Scores on the PHQ-9 (see Supplementary material available online at http://www.npjournal.org) were extracted from the electronic health record at baseline (before supplementation) and at follow-up visits after patients had consistently taken vitamin D for a minimum of 4 weeks and a maximum of 21 months. The primary outcome was the mean change in PHQ-9 scores, used to assess the clinical impact of the intervention on depressive symptoms. All data were deidentified before analysis.

Step 2

A standardized anticipatory guidance document was developed to support patient education. The resource summarized current evidence on the role of vitamin D in the management of depressive symptoms and was designed for use in clinical practice to promote adherence to supplementation protocols among individuals with confirmed vitamin D deficiency.

Measures

Depressive symptoms were assessed using the PHQ-9, a validated and widely used instrument with strong psychometric properties for screening and evaluating depression severity.[26] The PHQ-9 is reliable across diverse clinical settings and is publicly accessible, making it well-suited for use in outpatient psychiatric practice.[27] Normative data indicate that average scores among US adults range from 2.9 to 5.4. The minimal clinically important difference for the PHQ-9 has been reported to range from 2.5 to 4.5 points, with a 3-point reduction commonly accepted as a threshold for meaningful clinical improvement.[24]

Analysis

A paired-samples *t* test was used to compare mean PHQ-9 scores before and after intervention. Statistical significance and mean change were calculated to assess clinical impact.

Ethical Considerations

This retrospective electronic health record review was determined to be a quality improvement project and not human subjects research. All data were deidentified before analysis, and no patient contact occurred.

Results

The project included 39 patients, with a mean age of 40.46 (SD, 10.44; range 23-62 years). There were 6 men (15.4%) and 33 women (84.6%). The mean time from baseline to follow-up PHQ-9 administration was 10 months (SD, 6.8; range 1.15-21 months). A paired samples *t* test showed PHQ mean scores reduced significantly from 11.95 (SD, 5.25) at baseline to 5.92 (SD, 4.95) at follow-

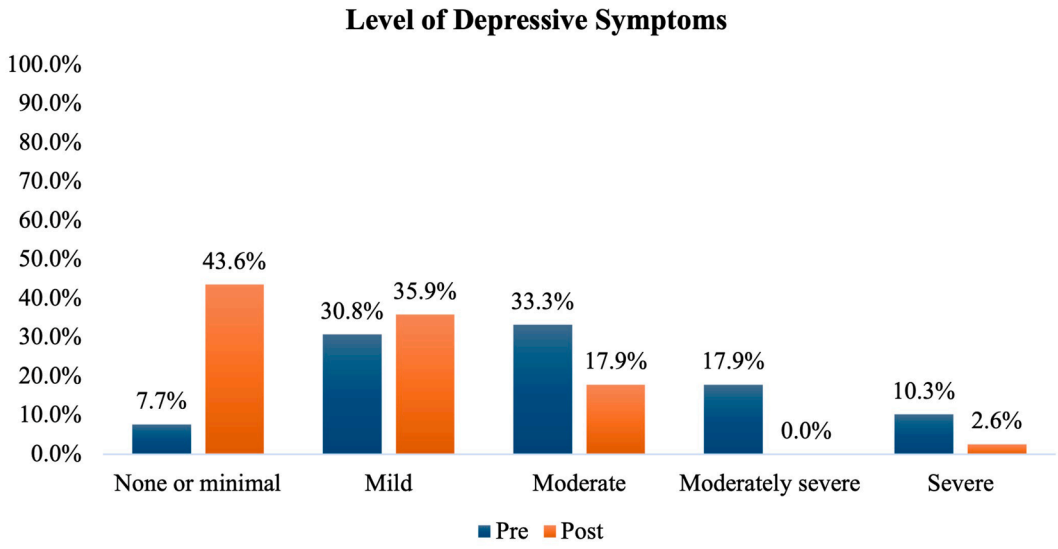

Figure. Depression scores before (pre) and after (post) vitamin D supplementation.

up ($t_{38} = 9.42$, $P < .001$). The mean reduction was 6.03 points (95% CI, 4.73-7.32 points; Cohen $d = 1.51$), indicating a large effect size. This reduction exceeded the goal set of 3.0 points and met the aim of the project.

The category of depressive symptoms was coded for each patient, and as shown in the Figure, there was a substantial improvement in the percentage of patients reporting none/minimal and mild depressive symptoms after the intervention and a decline in patients reporting moderately/severe or severe symptoms.

Time between PHQ administration and change in score was not associated (Spearman $\rho_{37} = -.201$, $P = .220$), suggesting that improvements were consistent across different time frames.

Discussion

Summary

This quality improvement project evaluated the impact of vitamin D supplementation on depressive symptoms in adult outpatients with documented vitamin D deficiency and treatment-resistant depressive disorders. The intervention of daily supplementation with 4,000 IU of vitamin D combined with standardized patient education resulted in a statistically and clinically significant reduction in PHQ-9 scores. The mean decrease of 6.03 points exceeded the predefined goal of a 3-point reduction and demonstrated a large effect size (Cohen $d = 1.51$), indicating meaningful improvement in symptom severity across the sample.

Interpretation

The findings of this project align with existing literature supporting the role of vitamin D in mood regulation. Prior studies have demonstrated that vitamin D supplementation may enhance serotonergic activity, reduce neuroinflammation, and promote neuroplasticity, particularly in individuals with baseline deficiency.[5] The observed improvements in PHQ-9 scores suggest that vitamin D may serve as a valuable adjunctive treatment in cases where traditional pharmacologic interventions have failed to produce adequate symptom relief. Importantly, the intervention was well tolerated,[13,15,17-23] and the educational component likely contributed to adherence and patient engagement. The lack of association between time to follow-up and symptom improvement further supports the consistency of the intervention's effect across varying durations.

Limitations

Several limitations should be considered when interpreting these findings. First, the retrospective design limits causal inference and is subject to documentation variability and selection bias. Second, the study did not control for confounding variables such as concurrent psychotherapy, medication changes, or lifestyle factors that may have influenced outcomes. Third, vitamin D metabolism is influenced by cofactors such as magnesium and vitamin K, which were not assessed in this review.[28,29] Additionally, genetic polymorphisms (eg, vitamin D receptor variants) may affect individual responses to supplementation and were not evaluated.[30] Other micronutrient deficiencies, including B_6, B_9, B_{12}, and zinc, may also modulate mood and interact with vitamin D pathways.[31] Finally, depression is a heterogeneous condition with diverse neurobiologic underpinnings; stratification by subtype or biomarker profile may yield more targeted insights in future studies.

Conclusion

This project demonstrated that vitamin D supplementation, when integrated into psychiatric outpatient care for individuals with depressive disorders and documented deficiency, can lead to clinically meaningful reductions in depressive symptoms. The intervention was simple, safe, accessible, and well tolerated, with a large effect size and consistent improvements across varying follow-up intervals. These findings support the incorporation of vitamin D screening and supplementation into routine mental health practice, particularly in regions with limited sun exposure and high prevalence of deficiency.

The development of a standardized educational resource further enhanced patient engagement and adherence, underscoring the importance of combining clinical interventions with patient-centered education. Although further prospective studies are needed to explore the role of genetic and nutritional cofactors in optimizing treatment response, this project contributes to the

growing evidence base supporting integrative approaches in psychiatric care.

CRediT authorship contribution statement

Janine Panker: Writing – review & editing, Writing – original draft, Project administration, Formal analysis, Data curation, Conceptualization. **Stacy Schoenbach:** Writing – review & editing, Validation, Methodology. **Pema Jessica Marcello:** Writing – review & editing, Validation, Methodology. **Julie Thompson:** Visualization, Methodology, Formal analysis. **Sean Convoy:** Writing – review & editing, Validation, Supervision, Resources, Methodology.

Declaration of Competing Interest

In compliance with standard ethical guidelines, the authors report no relationships with business or industry that may pose a conflict of interest.

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Supplementary Data

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